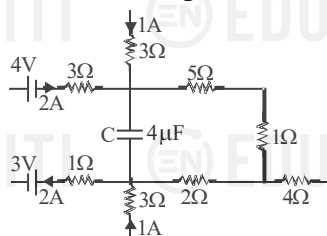
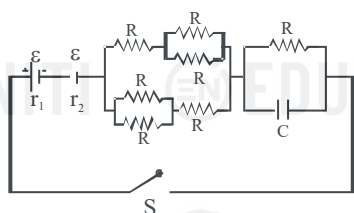


SUBJECTIVE TYPE

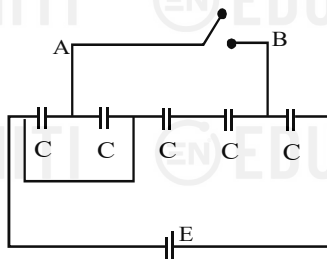
1. A part of circuit in steady state along with the currents flowing in the branches, is shown in the figure. Calculate the energy stored in the capacitor, ($C = 4\mu\text{F}$)



2. In the circuit shown in the figure, initially switch S is open and capacitor is uncharged. Internal resistances of the cells are r_1 and r_2 and their emf's are equal to $\mathcal{E}$. The potential difference across the cell of internal resistance r_1 becomes zero long time after closing the switch. Find the value of R in terms of other known physical quantities. All symbols have their usual meaning.



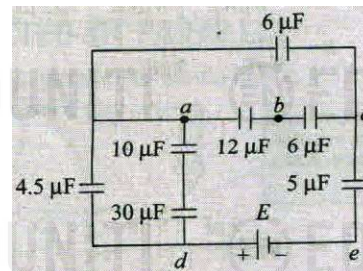
3. A parallel plate capacitor is filled with a dielectric whose relative permittivity varies with the applied voltage according to the law $\epsilon_r = aV$, where $a = 1$ per volt. Another capacitor having same dimensions but containing no dielectric is charged to potential 156V and connected in parallel to the first "non-linear" uncharged capacitor. Determine the final voltage across the capacitors.
4. Find the charge which flows from point A to B , when switch is closed.



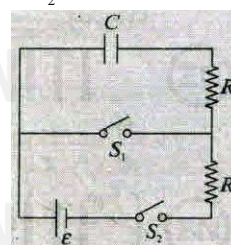
5. The plates of a parallel plate capacitor are separated by a distance $d = 1$ cm. Two parallel sided dielectric slabs of thickness 0.7 cm and 0.3 cm fill the space between the plates. If the dielectric constants of the two slabs are 3 and 5 respectively and a potential difference of 440V is applied across the plates, find
- the electric field intensities in each of the slabs.
 - the ratio of electric energies stored in the first to that in the second dielectric slab.

6. A charge $200\mu\text{C}$ is imparted to each of the two identical parallel plate capacitors connected in parallel. At $t = 0$, the plates of both the capacitors are 0.1 m apart. The plates of first capacitor move towards each other with a constant relative velocity 0.001 m/s and plates of second capacitor move apart with the same constant relative velocity. Find the current in the circuit.

7. In the circuit shown in fig., the potential difference between the points a and b is 4 V. Find the emf of the battery. Assume that before connecting the battery in the circuit, all the capacitors were uncharged.



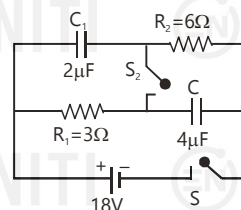
8. The capacitor shown in fig. has been charged to a potential difference of V so that it carries a charge CV with both the switches S_1 and S_2 remaining open. Switch S_1 is closed at $t = 0$. At $t = R_1 C$, switch S_1 is opened and S_2 is closed. Find the charge on the capacitor at $t = 2 R_1 C + R_2 C$



PARAGRAPH TYPE

Paragraph : Q9 to Q11

In the circuit shown, initially the switches are open and capacitors are uncharged. Switches S_1 and S_2 are closed simultaneously at $t = 0$.



9. Charge on capacitor C_2 in steady state is
- $12\mu\text{C}$
 - $24\mu\text{C}$
 - $48\mu\text{C}$
 - None of these
10. Now switch S_2 is opened after a long time interval then charge flow through the S_1 is
- $12\mu\text{C}$
 - $24\mu\text{C}$
 - $36\mu\text{C}$
 - $48\mu\text{C}$

11. In above question the amount of heat dissipated in resistors

- (a) 136 μJ (b) 272 μJ
(c) 68 μJ (d) 216 μJ

Paraph : Q12 to Q13

A sensor to find liquid level in a tank is a cylindrical capacitor of length ℓ with outer and inner conductor radii R_0 and R_i respectively. A liquid fills the tank up to height h , ($h \leq \ell$) from tank's bottom. The dielectric constant of liquid in the lower part is K_l and dielectric constant of vapour of liquid in the upper part is K_v .

12. When liquid fills the tank up to height h , the capacity of capacitor is

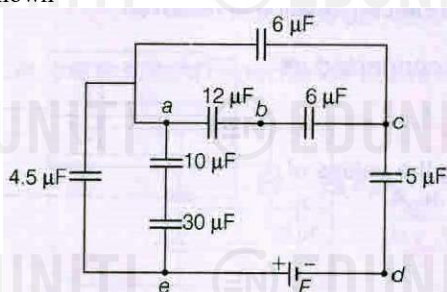
- (a) $\frac{2\pi\epsilon_0\ell}{\ln(R_0/R_i)} \left\{ (K_l - K_v) \frac{h}{\ell} + K_v \right\}$
(b) $\frac{2\pi\epsilon_0\ell}{\ln(R_0/R_i)} \left\{ (K_v - K_l) \frac{h}{\ell} + K_l \right\}$
(c) $\frac{2\pi\epsilon_0\ell}{\ln(R_0/R_i)} \left\{ (K_v + K_l) \frac{h}{\ell} + K_v \right\}$
(d) $\frac{2\pi\epsilon_0\ell}{\ln(R_0/R_i)} \left\{ (K_v + K_l) \frac{h}{\ell} + K_v \right\}$

13. If $\ell = 2\text{ m}$, $R_0 = 5\text{ mm}$, $R_i = 4.5\text{ mm}$, $K_l = 1.4$, $K_v = 1.0$, then, capacitance of capacitor when tank is fully filled is

- (a) $4.5 \times 10^{-9}\text{ F}$ (b) $7.5 \times 10^{-9}\text{ F}$
(c) $3.5 \times 10^{-9}\text{ F}$ (d) $1.5 \times 10^{-9}\text{ F}$

Paraph : Q14 to Q16

A combination of capacitors given is charged by a cell of emf E as shown



If it is given that the potential difference between points a and b is 4 V, then answer the given questions.

14. Potential difference between points a and c is

- (a) 4 V (b) 12 V
(c) 10 V (d) 8 V

15. Potential difference between points e and a is

- (a) 4 V (b) 12 V
(c) 10 V (d) 8 V

16. EMF E of the charging battery is

- (a) 46 V (b) 12 V
(c) 20 V (d) 18 V

Paraph : Q17 to Q19

Two parallel plate capacitors A and B have the same separation ($d = 8.85 \times 10^{-4}\text{ m}$) between the plates. The plate area of A and B are 0.04 m^2 and 0.02 m^2 respectively. A slab of a medium with dielectric constant $K = 9$ has same dimension such that it can exactly fill the space between the plates of capacitor B.

17. Dielectric slab is put between plates of A and it is charged to a potential of 110 V. Energy stored is

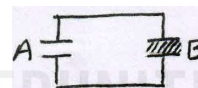
- (a) $1.21 \times 10^{-5}\text{ J}$ (b) $1.21 \times 10^{-3}\text{ J}$
(c) $1.21 \times 10^{-2}\text{ J}$ (d) $1.21 \times 10^{-6}\text{ J}$

18. Battery across A is disconnected and dielectric slab is removed slowly from A. Work done in removing the slab is

- (a) $4.84 \times 10^{-6}\text{ J}$ (b) $6.05 \times 10^{-5}\text{ J}$
(c) $1.21 \times 10^{-5}\text{ J}$ (d) $4.84 \times 10^{-5}\text{ J}$

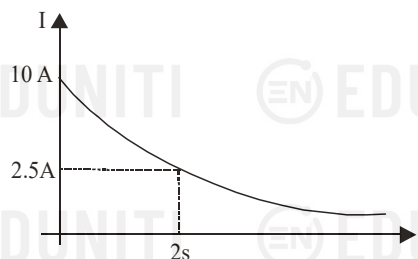
19. The same dielectric is now put in B and then A and B are connected as shown in fig. Energy stored in the system is

- (a) $1.21 \times 10^{-5}\text{ J}$
(b) $6.05 \times 10^{-5}\text{ J}$
(c) $4.84 \times 10^{-5}\text{ J}$
(d) $1.1 \times 10^{-5}\text{ J}$



MORE THAN ONE CORRECT

20. The figure shows, a graph of the current in a discharging circuit of a capacitor through a resistor of resistance 10Ω



(a) The initial potential difference across the capacitor is 100 volt.

(b) The capacitance of the capacitor is $\left(\frac{1}{10\ln 2}\right)\text{ F}$.

(c) The total heat produced in the circuit will be $\left(\frac{500}{\ln 2}\right)\text{ J}$.

(d) The thermal power in the resistor will decrease with time whose time constant is $\left(\frac{1}{2\ln 2}\right)\text{ second}$.

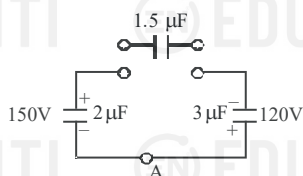


LEVEL UP - JEE ADVANCED 2022

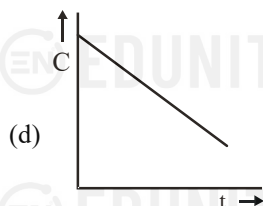
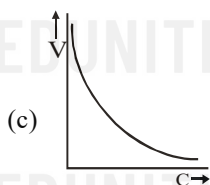
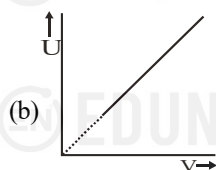
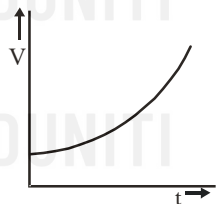
CAPACITORS

EDUNITI

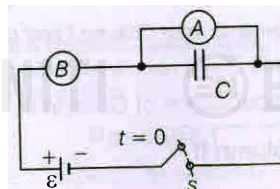
21. Two capacitors of capacitances $2\ \mu\text{F}$ and $3\ \mu\text{F}$ are charged to 150 volt and 120 volt respectively. The plates of capacitor are connected as shown in the figure. A discharged capacitor of capacity $1.5\ \mu\text{F}$ falls to the free ends of the wire. Then



- (a) charge on the $1.5\ \mu\text{F}$ capacitor is $180\ \mu\text{C}$
(b) charge on the $2\ \mu\text{F}$ capacitor is $120\ \mu\text{C}$
(c) charge flows through A from right to left
(d) charge flows through A from left to right.
22. A parallel plate capacitor A is filled with a dielectric whose dielectric constant varies with applied voltage as $K = V$. An identical capacitor B of capacitance C_0 with air as dielectric is connected to voltage source $V_0 = 30\text{V}$ and then connected to the first capacitor after disconnecting the voltage source. The charge and voltage on capacitor
- (A) A are $25C_0$ and $25V$ (B) A are $25C_0$ and $5V$
(C) B are $5C_0$ and $5V$ (D) B are $5C_0$ and $25V$
23. A parallel plate capacitor with square plates has a dielectric slab in it. The slab just fills the space inside the capacitor. The capacitor is charged by a battery and then battery is disconnected. Now at $t = 0$, the slab is pulled out slowly with constant speed in the direction parallel to one of the edges of the plates.. If at time t , capacitance of the capacitor is C , potential difference across is V and energy stored in it is U , then which of the following graphs are correct?



24. A capacitor of capacitance C is connected to two voltmeters A and B as shown in figure. The resistance of B is R and A is an ideal voltmeter of infinite resistance



The capacitor is initially uncharged and at $t = 0$, the switch is closed.

- (a) At $t = 0$, reading of A will be zero and that of B will be ε .
(b) During time interval $0 \leq t < \infty$, readings of B and A will change
(c) Readings of A and B will be equal at $t = RC \log_e 2$
(d) When $t \rightarrow \infty$, reading of A will be ε and that of B will be zero.